B.Sc. (Hons.)
CHEMISTRY Paper II
(Organic Chemistry)

1. An Organic compound A($C_{16}H_{16}$) on ozonolysis gives only one product B(C_8H_8O). Compound B on reaction with $NaBH_4$ gives a secondary alcohol (C). B and C gives positive iodoform test. C on oxidation with PCC give back B.Deduce the structures of A, B and C. Give equations and name of the reactions involved. (M-4)

2. Write down the product(s) obtained on dehydration of 3,3-dimethylbutan-2-ol.(M-1.5)

3. Justify the statement that aryl and vinyl halides show low reactivity towards nucleophilic substitution reaction compared to alkyl halides.(M-1.5)

4. Why is nitration of toluene much faster compared to nitration of nitrobenzene? Name the product(s) formed in each case.(M-2)

5. Which one is more acidic ortho or p-nitrophenol and why? (M-1.5)

6. Account for aromaticity observed in 1,3,5-cycloheptatrienyl cation but not in 1,3,5-cycloheptatriene.(M-1)

7. o-Bromomethoxybenzene on reaction with sodamide in liquid ammonia, does not give o-methoxyaniline. What is the product obtained in this reaction? Explain with mechanism. (M-1.5)

8. What are the products obtained when 2-chloro-1-phenylpropane undergoes dehydrohalogenation? Which is the major product and why? (M-1.5)

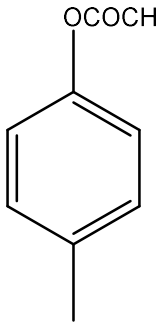
9.Arrange and explain the order of nucleophilicity of halide ions(F, Cl,Br,I) in water.

10.Discuss the stereochemistry of addition of bromine to E-But-2-ene.(M-2)

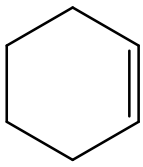
11. Carry out the following conversions :-
(a) Acetaldehyde to n-butyl alcohol
(b) Propyne to tert-butyl alcohol
(c) Benzene to m-bromophenol
(d) Isopropyl chloride to n-propyl chloride (M-4×1.5)

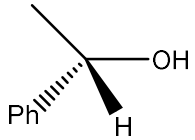
12. Complete the following reactions (with mechanism) :-

1. $(CH_3CH_2)_3Br \xrightarrow{25^{o}\,C/\,CH_3OH}$?

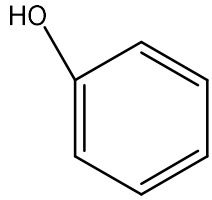
2. 

$\xrightarrow[\triangle]{AlCl_3}$?

3. 
$\xrightarrow{NBS}$?

4. 

$\xrightarrow{SOCl_2}$?

5 

$+CO_2 \xrightarrow[H^+]{NaOH\text{-}100\ ^{o}C}$?

13.How many moles of HIO_4, are consumed by one mole of 2,4-dimethyl-2,3,4-hexanetriol? Write down the products obtained in this reaction. (M-1.5)

14.Write down the mechanism of formation of phenol from cumene. (M-1.5)
15.Whether o or m- chloronitrobenzene undergoes faster nucleophilic substitution and why? (M-2)
16.Give a detailed account of Hofmann elimination and compare it with Satyzeff Elimination.

17.How you will distinguish the following, Give visible test:
(I) Chlorobenzene and Chloroethane
(II) Ethanol and phenol
(III) Diethyl ether and ethanol
(IV) Phenol and p-cresol
(V) Phenol and benzyl Alcohol
(VI) Primary alcohol, secondary alcohol and tertiary alcohol. (M-3)

18.Explain the following:
(i) '-R(withdrawing)' group at ortho and para position facilitates aryl halides towards Aromatic Nucleophillic Substitution Reaction.
(ii) Neopentyl halide are notoriously slow in nucleophilic substitution whatever the experimental conditions are
(iii) Which of the following 2 synthesis is preferred for tertiary butyl ether

1. $(CH_3)_3C\text{-}O^- + C_2H_5Br \longrightarrow$

2. $(CH_3)_3C\text{-}Br + C_2H_5O^- \longrightarrow$

IV) Compare the reactivity of chlorobenzene and 2,4 – dinitrochlorobenzene towards NaOH.

V) Nitration of bromobenzene is much faster than bromination of nitrobenzene.
VI) Methanol has higher boiling than methyl chloride.
VII) Methanol is more acidic than water whereas water is more acidic than ethanol.
VIII) Solvation and hydrogen bonding stabilize the negative charge on oxygen, which is correlate to pKa value of alcohols.
IX) We can not prepare aldehydes from direct oxidation of alcohols using strong oxidising agents (conc.HNO_3 or $KMnO_4$)
(M-1.5)

19. How will you carry out the following conversions :
(i) Phenol to Asprin
(ii) Propanal to 2-butanol
(iii) Propanoic acid to propanol
(iv) Butanone to Butanol
(v) Acetone to 2-methyl-2-butanol
(vi) n-propanol to butanamide (M-2)

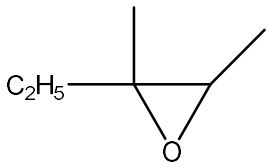
20.What happens when:
(i) Tertiary alcohols are heated with conc.sulphuric acid.
(ii) Glycerol is treated with periodic acid.
(iii) Chloro benzene is heated with Soda amide.
(iv) 2,3-Dimethyl butan-2,3-diol is treated with H_2SO_4. (M-1.5 each)

21.Write a short note on the followings: (M-3 each)
(a) Pyridinium chloroformate (PCC) or DMP
(b) Fries Rearrangement
(c) Oppenauer Oxidation
(d) Pinacol-pinacolone reaction.
(e) Reimer-Tiemann Reaction.
(f) Houben-Hoesch reaction.
(g) Claisen Rearrangement/3,3 sigmatrophic rearragement.

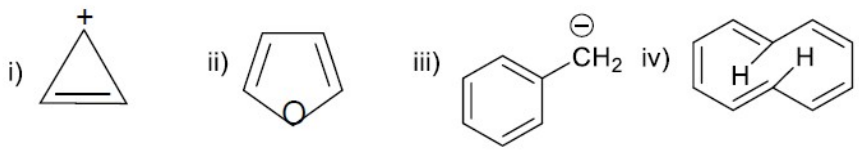
22.Write the structure of alcohol formed from $(CH_3)_2C{=}CH{-}CH_3$ on hydroboration - oxidation, give the mechanism involved.

23.What products are formed when Anisole is heated with HI? Explain with the help of mechanism.

24.Write the reaction sequence involved in the ring opening with methanol in presence of acid.Explain the formation of different products on reaction with sodium methoxide.



25.Classify the following compounds as aromatic, anti-aromatic and non-aromatic with suitable explanation:



26.Addition of hydrogen chloride to 3,3-dimethyl but-1-ene yields a mixture of 2-chloro-3,3- dimethylbutane and 2-chloro-2,3-dimethylbutane. Explain.

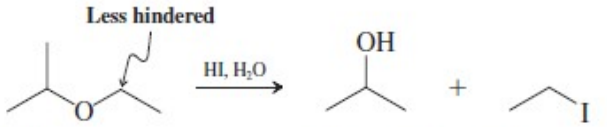
27.Explain with mechanism that when Isobutyl chloride reacts with benzene using anhydrous $AlCl_3$ the product obtained is not Isobutyl benzene but t-butyl benzene.

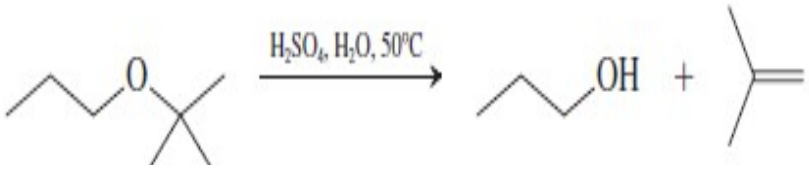
28. Explain the formation of following products in ether.

Primary Ether Cleavage with HBr

$$\underset{\text{Ethoxyethane}}{CH_3CH_2OCH_2CH_3} \xrightarrow{HBr} \underset{\text{Bromoethane}}{CH_3CH_2Br} + \underset{\text{Ethanol}}{CH_3CH_2OH}$$

Primary-Secondary Ether Cleavage with HI: S_N2 at Primary Center



2-Ethoxypropane **2-Propanol** **Iodoethane**

Primary-Tertiary Ether Cleavage with Dilute Acid: S_N1 and E1 at Tertiary Center